Model Summary

Data level

$$y_{ijk} \sim \text{Bernoulli}\left(p_{ijk}\right)$$
$$p_{ijk} = \Phi\big(\beta_j\left(s_{ijk} + \theta_{ij}\right)\big)$$

Measurement level

$$\theta_{ij} = \mu_j + \sigma_j\,\theta^*_{ij}, \qquad \text{Var}(\theta^*_{ij}) = 1, \qquad j = 1, \dots, 6$$

The scale of θ_{ij} is set by the stimulus units of each task and is therefore arbitrary across tasks. The loadings are defined on the standardized abilities θ^*_{ij}, so that equal loadings mean an equal proportion of variance in every task, independent of units; σ_j absorbs the scale of each task.

$$\gamma^*_i = \Lambda\,\eta_i + \varepsilon_i$$
$$\gamma^*_i = \left(\theta^*_{i1}, \theta^*_{i2}, \theta^*_{i3}, \theta^*_{i4}, \theta^*_{i5}, \theta^*_{i6}, c^*_{i1}, c^*_{i2}, c^*_{i3}, c^*_{i4}\right)'$$
$$\Lambda = \begin{pmatrix} \lambda_1 & \lambda_{12} & 0 & \lambda_4 & 0 & 0 & 0 \\ \lambda_1 & \lambda_{22} & 0 & 0 & \lambda_5 & 0 & 0 \\ \lambda_1 & \lambda_{32} & 0 & 0 & 0 & \lambda_6 & 0 \\ \lambda_1 & 0 & \lambda_{43} & \lambda_4 & 0 & 0 & 0 \\ \lambda_1 & 0 & \lambda_{53} & 0 & \lambda_5 & 0 & 0 \\ \lambda_1 & 0 & \lambda_{63} & 0 & 0 & \lambda_6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{87} \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{97} \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{10,7} \end{pmatrix}, \quad \eta_i = \begin{pmatrix} \eta_{i1} \\ \eta_{i2} \\ \eta_{i3} \\ \eta_{i4} \\ \eta_{i5} \\ \eta_{i6} \\ \eta_{i7} \end{pmatrix}, \quad \varepsilon_i \sim N(\mathbf{0},\, \text{diag}(\delta^2_1, \dots, \delta^2_{10}))$$

The general loading λ_1 is shared by all six perception tasks, and the stimulus loadings $\lambda_4, \lambda_5, \lambda_6$ are shared by the temporal and spatial version of the same stimulus; the masking loadings are free. Without these equalities the loadings are not identified: each stimulus factor has only two indicators, and the general loadings of the temporal tasks can be traded against those of the spatial tasks.

Structural level

$$\eta_i = \mathbf{B}\,\eta_i + \zeta_i, \qquad \zeta_i \sim N(\mathbf{0}, \mathbf{I}_7)$$
$$\mathbf{B} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ b_1 & b_2 & b_3 & b_4 & b_5 & b_6 & 0 \end{pmatrix}$$

$$\left(\lambda_{12} + \lambda_{22} + \lambda_{32}\right) b_2 + \left(\lambda_{43} + \lambda_{53} + \lambda_{63}\right) b_3 = 0$$
$$\lambda_4\, b_4 + \lambda_5\, b_5 + \lambda_6\, b_6 = 0$$

A link of cognition with the general factor is indistinguishable from an equal link with all stimulus (or all masking) factors. The two constraints remove this ambiguity by contrast coding: the masking and the stimulus contributions to the task–cognition correlations each sum to zero, so b_1 carries the link common to all six tasks and $b_2, \dots, b_6$ capture deviations from it. b_3 and b_6 follow from the constraints; b_1, b_2, b_4, b_5 are free.

Variance decomposition of perceptual task j

$$\mathrm{Var}(\theta_{ij}) = \sigma_j^2 \Big(\underbrace{\Lambda_{j1}^2}_{\text{General}} + \underbrace{\Lambda_{j2}^2 + \Lambda_{j3}^2}_{\text{Masking}} + \underbrace{\Lambda_{j4}^2 + \Lambda_{j5}^2 + \Lambda_{j6}^2}_{\text{Stimulus}} + \underbrace{\delta_j^2}_{\text{Unique}} \Big) = \sigma_j^2$$

Relation to cognitive performance

$$v = \mathrm{Var}(\eta_{i7}) = 1 + \sum_{l=1}^{6} b_l^2$$

$$r_l = \mathrm{Corr}(\eta_{il},\, \eta_{i7}) = \frac{b_l}{\sqrt{v}}, \qquad l = 1, \dots, 6$$

$$R_G^2 = r_1^2, \qquad R_M^2 = r_2^2 + r_3^2, \qquad R_S^2 = r_4^2 + r_5^2 + r_6^2$$

Bayes factors

$$BF_{10}^G = \frac{p(b_1 = 0)}{p(b_1 = 0 \mid \mathbf{Y})}, \qquad BF_{10}^M = \frac{p(b_2 = 0)}{p(b_2 = 0 \mid \mathbf{Y})}, \qquad BF_{10}^S = \frac{p(b_4 = b_5 = 0)}{p(b_4 = b_5 = 0 \mid \mathbf{Y})}$$

Because of the constraints, $b_2 = 0$ implies $b_3 = 0$, and $b_4 = b_5 = 0$ implies $b_6 = 0$.

Result 1: Variance decomposition

Task	General	Masking	Stimulus	Unique
TPi	0.31 [0.19, 0.45]	0.04 [0.00, 0.15]	0.37 [0.22, 0.51]	0.28 [0.14, 0.43]
TLetter	0.31 [0.19, 0.45]	0.21 [0.03, 0.47]	0.25 [0.10, 0.41]	0.22 [0.02, 0.45]
TGabor	0.31 [0.19, 0.45]	0.15 [0.01, 0.44]	0.20 [0.05, 0.38]	0.33 [0.06, 0.56]
SPi	0.31 [0.19, 0.45]	0.16 [0.02, 0.35]	0.37 [0.22, 0.51]	0.16 [0.02, 0.33]
SLetter	0.31 [0.19, 0.45]	0.22 [0.04, 0.46]	0.25 [0.10, 0.41]	0.21 [0.02, 0.45]
SGabor	0.31 [0.19, 0.45]	0.14 [0.01, 0.38]	0.20 [0.05, 0.38]	0.34 [0.10, 0.56]
Average	**0.31 [0.19, 0.45]**	**0.15 [0.08, 0.24]**	**0.27 [0.18, 0.38]**	**0.26 [0.18, 0.35]**

Result 2: Relation to cognitive performance

Factor	r	H_0	BF_{10}
General visual	0.64 [0.40, 0.83]	$b_1 = 0$	13,840.56
Temporal masking	-0.20 [-0.45, 0.05]	$b_2 = 0$	0.79
Spatial masking	0.16 [-0.04, 0.37]		
Pi	0.05 [-0.17, 0.26]	$b_4 = b_5 = 0$	0.56
Letter	-0.28 [-0.51, -0.03]		
Gabor	0.24 [-0.07, 0.54]		